OMNITOME PUBLISHING | SYNTHETIC ARCHIVE

The Curious World of Cats

OCR Test Plates | October 2026



1. Craniodental Specialization in *Felis catus*

Across all forty-one extant species of the family Felidae, the shortening of the facial rostrum and reduction of the post-carnassial molar arcade reflect an uncompromising evolutionary commitment to vertebrate predation. In the adult domestic cat (*Felis catus*) and its Near Eastern progenitor (*Felis lybica*), the permanent dental formula is strictly conserved at thirty teeth:

> Dental Formula: I 3/3, C 1/1, P 3/2, M 1/1 (× 2 = 30 teeth)

The upper third premolar (P^3) and lower first molar (M_1) form the sectorial carnassial shear. Because the feline temporomandibular joint is a tight cylindrical hinge encased by deep bony processes, lateral chewing excursion is eliminated, concentrating masseter and temporalis force along a vertical scissor plane capable of severing avian tendons and rodent cervical vertebrae.

2. Enzymatic Rigidity and the Sweet-Receptor Pseudogene (ψ*Tas1r2*)

Unlike facultative carnivores such as the domestic dog (*Canis familiaris*), the feline liver maintains constitutively high activity of transaminases and deaminases regardless of dietary protein intake. Furthermore, a 247-base-pair microdeletion in exon 3 of the *Tas1r2* gene renders the mammalian T1R2/T1R3 sweet taste receptor non-functional (Li et al., 2005), while leaving the T1R1/T1R3 umami receptor tuned to L-amino acids and purine nucleotides at prey body temperature (35°C to 38.5°C).

FIELD MARGIN NOTE

Key Diagnostic Rule: Only Clade IV (*Felis lybica*) exhibits the calm hypothalamic-pituitary-adrenal (HPA) stress profile required for Neolithic granary commensalism; Clade I (European *F. silvestris*) remains hyper-defensive even when hand-reared.

Palaeogenetic Clade Architecture

Mitochondrial *ND5/ND6* sequencing across 979 wild and domestic felids (Driscoll et al., 2007) partitions *Felis silvestris sensu lato* into five phylogeographic clades. Ancient DNA from 209 archaeological remains (Ottoni et al., 2017) confirms two dispersal waves: Wave 1 (IV-A) from Neolithic Anatolia (~8000–4400 BCE) and Wave 2 (IV-C) from Classical Egypt (~1500 BCE–500 CE).

Clade	Taxonomic Subspecies	Geographic Distribution	Domestication Contribution
Clade I	*Felis silvestris silvestris*	Europe & Scotland	None (occasional hybrid introgression)
Clade II	*Felis lybica cafra*	Southern Africa	None (diverged ~131,000 years ago)
Clade III	*Felis lybica ornata*	Central Asia & India	None (secondary trade admixture only)
Clade IV	*Felis lybica lybica*	Near East & North Africa	**Sole maternal ancestor (IV-A to IV-E)**
Clade V	*Felis bieti*	Tibetan Plateau Margin	None (distinct alpine lineage)

Table OCR-A2.1: Comparative phylogeography of the five wildcat clades and their maternal contribution to modern Felis catus.